

Statewide Marine Habitat Map 2023

Title:	Statewide Marine Habitat Map 2023
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Resource Name:	VIC_MARINE_HABITAT_MAP_2023
Publication Date:	28-11-2023
Spatial Extent:	General - Victoria
Owner:	Department of Energy, Environment and Climate Action
Custodian:	Department of Energy, Environment and Climate Action
Data access:	Unclassified
Jurisdiction:	Victoria
Abstract:	The Statewide Marine Habitat Map 2023 was developed by DEECA applying novel machine learning methods that model and predict habitat distributions as well as a mosaic of former mapping products (listed below). The Statewide map represents 24 marine and coastal habitats complexes at Level 3, Victoria's Combined Biotope Classification Scheme (CBiCS) described by Edmunds and Flynn (2015, 2018; 2021). The final map comprises of 83% its area from predictive modelling, with the remaining 17% of area from synthesised existing habitat maps. Predictive Model: A total of 32,998 habitat survey sites (ground-truth records) were used within the model, along with 28 environmental properties mapped at a 10m resolution (including a Digital Elevation Model DEM (VCDEM2021), computed benthic terrain characteristics (toolkit: Walbridge et al. 2018), Chlorophyl a (IMOS 2000a), Sea Surface Temperature SST (IMOS 2000a), Net Primary Productivity NPP (IMOS 2000b), Sediments (Geoscience Australia; Li et al. 2011a,b,c), waves (Liu et al. 2022). To predict the distribution of habitats across Victorian waters the powerful and flexible Random Forest machine learning algorithm was applied. Random Forest is an ensemble model using bagging as the ensemble method and decision trees as the individual model (Breiman 2001). The modelling produced an accuracy (Out-of-bag) of 89%. Map Synthesis: A mosaic of former mapping products that provided higher resolution mapping by aerial imagery, field observations and high-resolution modelling were integrated into the map, classifying habitat according to the CBiCS habitat classification scheme at level 3. Assessed and synthesised maps and citations include: Corangamite Coast Marine Habitat December 2009 (ANZVI0803005530); East Gippsland Marine Habitats November 2009 (ANZVI0803003974); Discovery Bay Marine National Park habitat mapping 2006 (ANZVI0803004053); Portland Coastal Habitats (ANZVI0803004236) ; Corner Inlet Mapping Marine National Park North and South 2004 (ANZVI0803004051) ; Merri Marine Sanctuary 2004 (ANZVI0803004058); Western Port Bay Biotope Mapping Fathom Pacific (2016) CBiCS-Mapping. Central Victoria Coastal Habitats (ANZVI0803004135); Mallacoota Coastal Habitats (ANZVI0803004235); Western Port Rhodolite (ANZVI0803005430) & Western Port Biogenic Reefs; Port Phillip Bay Habitat Map 2021 (ANZVI0803009278); Saltmarsh and Mangrove Habitats; DELWP 2021 Statewide Marine Habitat Map 2021 (ANZVI0803009286) and relevant citations: Ball (1999), Ball et al. (2010). Ball & Blake (2007a), Ball & Blake (2007b), Blake and Ball (2001), Blake et al. (2013), Boon et al. (2011), Cohen et al (2000), Deakin Marine Mapping (Zavalas, R et al. 2018), DELWP (1994), Edmunds & Flynn (2015), Fathom Pacific (2020), Ford et al (2016), GeoHab Victoria Estuaries Geomorphology (2010),

Ierodiconou 2007, Ierodiconou et al. 2018, Mazor et al. (2021), Monk et al. (2011), Poore (1992), Roob and Ball (1997), Victoria Department of Transport (1999), Young et al. 2022, Zavalas, R et al. 2018.

Applications: The Statewide Marine Habitat Map 2023 provides broad habitat complexes across the state and provides greater knowledge of the ecological diversity across Victoria's waters. The map should be used at broad scales of >25 m, and where information of larger habitat complexes is needed. This work can support the management of large-scale habitats, their condition, marine spatial planning, strategic management prospect (SMP), FeAST risk assessments, and other broad scale applications to support management decisions across Victoria. The habitat model and resulting map provides an updated broad-scale habitat map across Victoria's state waters and provides a baseline for future data to build upon.

Full Methodology:

Citation: Mazor, T., Watermeyer, K., Hobley, T., Grinter, V., Holden, R., MacDonald, K. and Ferns, L. (2023). Statewide Marine Habitat Map. Habitat Complex Modelling Method (CBiCS Level 3). The State of

Purpose:

The statewide marine habitat map provides broad habitat complexes across the state and provides greater knowledge of the ecological diversity across Victoria's waters. The map should be used at broad scales of >25 m, and where information of larger habitat complexes is needed. This work can support the management of large-scale habitats, their condition, marine spatial planning, strategic management prospect (SMP), FeAST risk assessments, and other broad scale applications to support management decisions across Victoria. The state habitat map addresses the Marine & Coastal Strategy 2022, building knowledge to improve the condition and ecological connectivity of habitats, and protecting Victoria's environment - Biodiversity 2037, improving our knowledge of habitats and their extent to support their protection and restoration.

Data Currency Information:

Data Set Status:

Completed

Data Collection:

Collection Period:

2023-01-23 - 2023-01-23

Update Frequency:

As needed

Dataset Information:

Dataset Source:

Dataset Source: PPB Habitat model 2021 provides an updated broad-scale habitat map across Port Phillip Bay, which has applied new machine learning techniques. Previous mapping of Port Phillip Bay has provided insight into habitats across the region, with the latest map produced in 2016 (Edmunds & Flynn 2018). Since then, additional benthic survey data and data on environmental predictors has developed which increases the capacity to improve model habitat distributions and extent.

The Open Coast statewide marine habitat map (excluding PPB) developed by Deakin Marine Mapping lab using a range of techniques including digitisation of polygons and has been compiled using a compilation of various datasets including Boon et al. (2011). Deakin Marine Mapping (Young M et al, 2018). GeoHab Victoria Estuaries Geomorphology (2010). Deakin Marine Mapping (Zavalas, R et al. 2018), DELWP wetlands (1994), Monk et al 2011, Roob and Ball (1997), Ford et al 2016, Edmunds and Flynn (2015), Blake and Ball (2001), Poore (1992), Cohen et al (2000). The map has later been

reclassified to CBICS habitat classification scheme at broad-scale level 2 and 3.

Dataset Originality: Primary & Derived

Dataset Processing Details:

Positional Accuracy:

Modelled to coordinate system: VICGRID94 Due to modelling and digitisation uncertainties the maps should be used at broad scales of over 25m for decision making. Scale range on CoastKit available at 1:100.000

Attribute Accuracy:

Modelled estimation of habitat distribution using best available data up to 2021 see report.

Logical Consistency:

Not applicable

Completeness:

Victoria's marine environment and coastal habitats. Some small data gaps as a relict of the modelling process and availability of environmental predictor data. Completeness Verification: Habitat distribution model prediction using random forest machine learning tools from 32,998 ground-truthed habitat records and other digitised compiled datasets.

Access:

Constraints:

Publicly available under Creative Commons Licence . Citation for use: Mazor, T., Watermeyer, K., Hobley, T., Grinter, V., Holden, R., MacDonald, K. and Ferns, L. (2023). Statewide Marine Habitat Map. Habitat Complex Modelling Method (CBiCS Level 3). The State of Victoria Department of Energy, Environment and Climate Action.

Stored Data Format:

Available Format Type:

Digital

Search:

Search Word:

- Biota
- Oceans

Further Information:

Supporting Documentation:

Related Documents: None

History:

Stages:

Publication

2023-02-01T01:09:51

Last Updated:

Date:

Related Datasets:

No related datasets

Additional Metadata URL:

<http://www.delwp.vic.gov.au/vicmap>

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